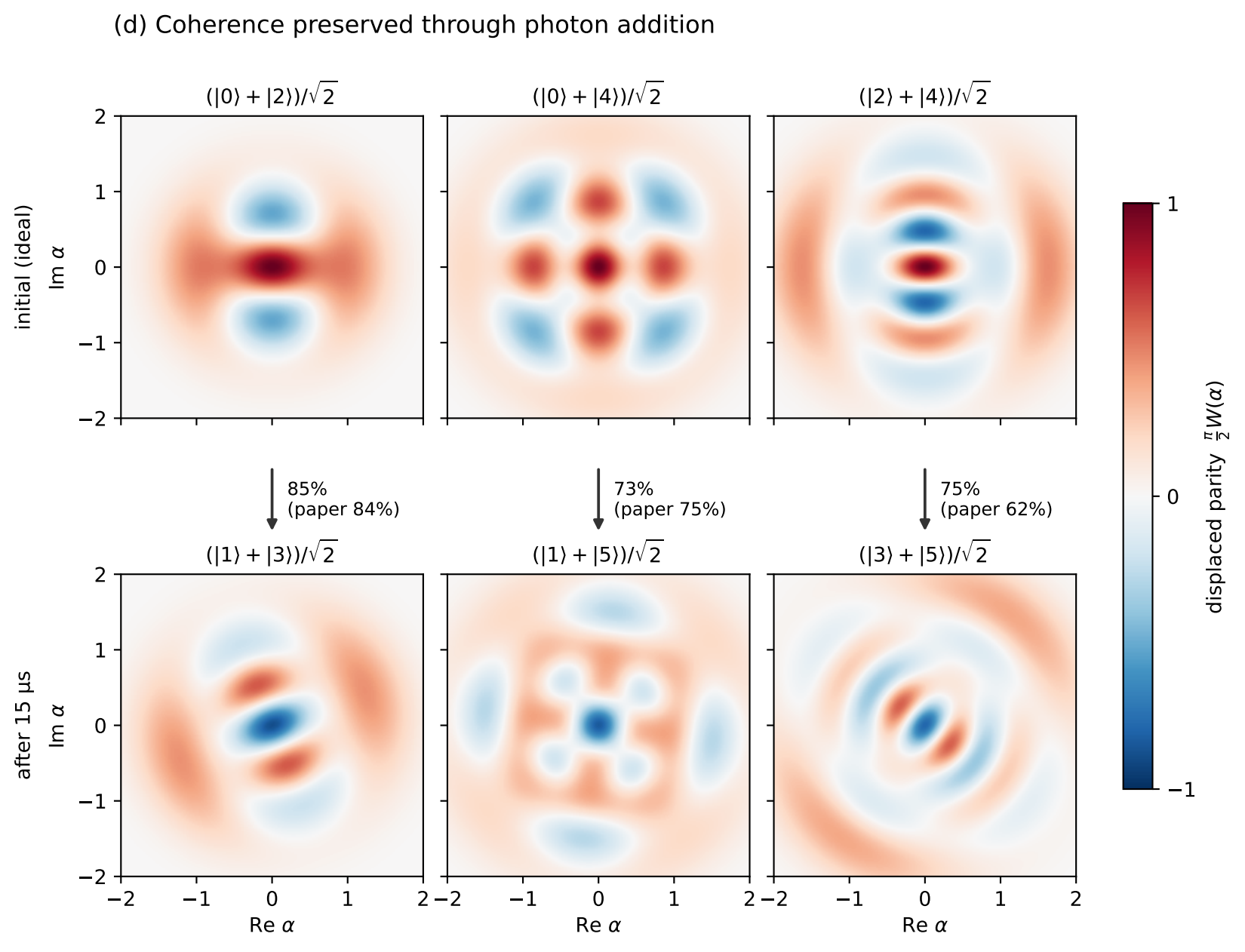

(d) Coherence preserved through photon addition
$(|0\rangle + |2\rangle)/\sqrt{2}$
$(|0\rangle + |4\rangle)/\sqrt{2}$
$(|2\rangle + |4\rangle)/\sqrt{2}$
initial (ideal)
Im α
85%
(paper 84%)
73%
(paper 75%)
75%
(paper 62%)
$(|1\rangle + |3\rangle)/\sqrt{2}$
$(|1\rangle + |5\rangle)/\sqrt{2}$
$(|3\rangle + |5\rangle)/\sqrt{2}$
after 15 µs
Im α
Re α
Re α
Re α
displaced parity $\frac{\pi}{2}W(\alpha)$
1
0
−1